

WhereTheWaterFlows.jl: Hydrological flow routing on digital elevation models in Julia

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Summary

WhereTheWaterFlows.jl (WWF) is a Julia package for computing water flow paths and drainage basins on digital elevation models (DEMs) or other hydropotential surfaces. Given a DEM, the package determines *where the water flows* across the landscape, accumulating upslope area and delineating catchment basins. Besides this classic, deterministic routing on surface topographies, WWF also includes routines to route water in subglacial settings as well as uncertainty quantification tools based on a Monte Carlo approach.

Key outputs include the flow-direction field, the upslope contributing area or discharge, the depression-filled DEM, a labelled catchment-basin map, and the location of all sinks. Plotting functions for result visualisation are included. The package is registered in the Julia General registry, has a full test suite and complete [online documentation](#).

WWF implements the widely-used D8 single-flow-direction algorithm, which routes water from each grid cell to its lowest-elevation neighbour among the eight surrounding cells, together with a breach-type depression-filling algorithm to handle closed depressions that would otherwise trap flow (O'Callaghan & Mark, 1984). The underlying graph traversal follows the O(n) recursive strategy of Braun & Willett (2013), giving linear scaling with the number of grid cells. WWF performs comparably to or better than existing routing software, as shown in our [benchmarks](#).

Statement of need

Hydrological flow routing on DEMs is a fundamental operation in geosciences: it underpins catchment delineation, runoff modelling, subglacial hydrology, and landscape-evolution studies. Software implementations of flow routing date back to O'Callaghan & Mark (1984) and, nowadays, many implementations exist. WWF is the first native Julia package for this task.

WWF is suitable for any domain where DEM-based flow analysis is required. However, the package has been designed with glaciological applications in mind where it has been used in six publications to date, see section [Research impact](#). The glaciological focus required the implementation of glacier specific aspects, such as Shreve potential routing (Shreve, 1972) in the Subglacially submodule, as well as the development of uncertainty quantification via Monte Carlo methods (the Randomly submodule). The latter aspect is needed in glaciology as uncertainties in glacier bed DEMs and Shreve-potential are large and thus warrant quantification.

State of the field

Several mature tools exist for hydrological flow routing on DEMs, examples include:

- 39 ■ **Geomorphometry.jl** is a Julia package providing many geomorphic analysis tools, among
40 them flow routing. ([Pronk & Schouten, 2026](#))
 - 41 ■ **GRASS GIS** (command line/C/Python/QGIS), the component `r.watershed` implements
42 several water routing and related algorithms. ([GRASS-Team et al., 2026](#))
 - 43 ■ **TauDEM** (command line/ArcGIS) provides command-line tools designed for large-scale
44 catchment analysis. ([Tesfa et al., 2011](#))
 - 45 ■ **TopoToolbox** (MATLAB/Python) provides widely used functionalities for topographic
46 analysis including flow routing. ([Schwanghart & Scherler, 2014](#))
 - 47 ■ **Whitebox Tools** (Rust/Python) provides an extensive geomorphometry toolkit as a Rust
48 library with Python bindings and unofficial [Julia bindings](#). ([Lindsay, 2016](#))
 - 49 ■ **RichDEM** (Python/C++) offers a broad set of flow-routing methods. ([Barnes et al.,](#)
50 [2014](#))
- 51 To our knowledge, none of the existing routing software packages implement specialised
52 subglacial routing functionality or have uncertainty quantification capabilities.

53 Software design

54 The central function is `waterflows(dem)`, which accepts a two-dimensional array of elevation
55 values and returns a named tuple of outputs such as upstream area and catchments. Boundary
56 conditions (open or closed edges, fixed sinks) and other options can be specified via keyword
57 arguments. A novel feature of WWF is the feedback-function (via the `feedback_fn` argument
58 of `waterflows`) which is called on each cell before routing its water downstream, enabling
59 feedbacks to occur. This allows for extensive customisations, as demonstrated in the provided
60 examples (see `examples/` directory) on sediment transport and subglacial melt calculations.

61 For subglacial routing, the main function is `waterflows_subglacial(surfdem, beddem, dx,`
62 `f)` with surface and bed DEMs, grid spacing and water pressure as flotation fraction as inputs.
63 In the subglacial setting, the `beddem` and `f` input fields have large uncertainties associated with
64 them. This spurred the development of the Monte Carlo based uncertainty quantification in
65 WWF (which can be used independently of subglacial routing). The main function there is
66 `map_mc(model, sample, reduce!, n)` which runs the `model` function (i.e. the water routing)
67 on `n` samples of surface, bed and flotation fractions fields and reduces the results with `reduce!`
68 (to avoid huge data volumes). The uncertainty in the input fields is modelled using Gaussian
69 Random Fields, which provide spatially correlated random noise. The random fields are
70 generated using a FFT-based method ([Räss et al., 2019](#)). However, the software design allows
71 to couple more sophisticated, external geostatistical models to generate the uncertainty fields.

72 From the outset, WWF has been developed with subglacial applications in mind where
73 uncertainties in inputs are large and would likely dominate over routing algorithm choices.
74 Therefore, the simplest routing algorithm, D8, was deemed sufficient and implemented.
75 Incidentally, the ensemble of flow paths resulting from stochastic routing exhibits an effectively
76 diffusive flow (which is what more sophisticated routing schemes, such as D-inf provide).
77 Although subglacial routing was the motivation to develop WWF, the main WWF module and
78 the `Randomly` submodule were designed, again from the outset, to be usable and useful for
79 general settings. Making WWF useful for general settings also motivated our choice to keep
80 the subglacial and stochastic functionality tucked away in submodules, which makes it clear
81 that these are optional features and that WWF can be used without them. The API of WWF
82 is intentionally kept simple and is based around using generic vectors and arrays as inputs
83 and no custom datatypes are defined (with the exception of the `Uncertainty` struct). This
84 makes WWF interoperable with many grid based Julia packages, in particular georeferenced
85 raster data of the standard packages [Rasters.jl](#) and [GeoArrays.jl](#) can be used and leads to
86 georeferenced WWF outputs (except for `Randomly` workflows which currently return normal
87 arrays).

Example

The code and figure below show an example of a drainage network calculation on a small synthetic DEM:

```
using WhereTheWaterFlows, CairoMakie
n = 200
x = y = range(-π, π, length=n)
dem = sin.(x) .* cos.(y') .+ 0.05 * rand(n, n)

out = waterflows(dem)
plt_area(x, y, out.area)
```

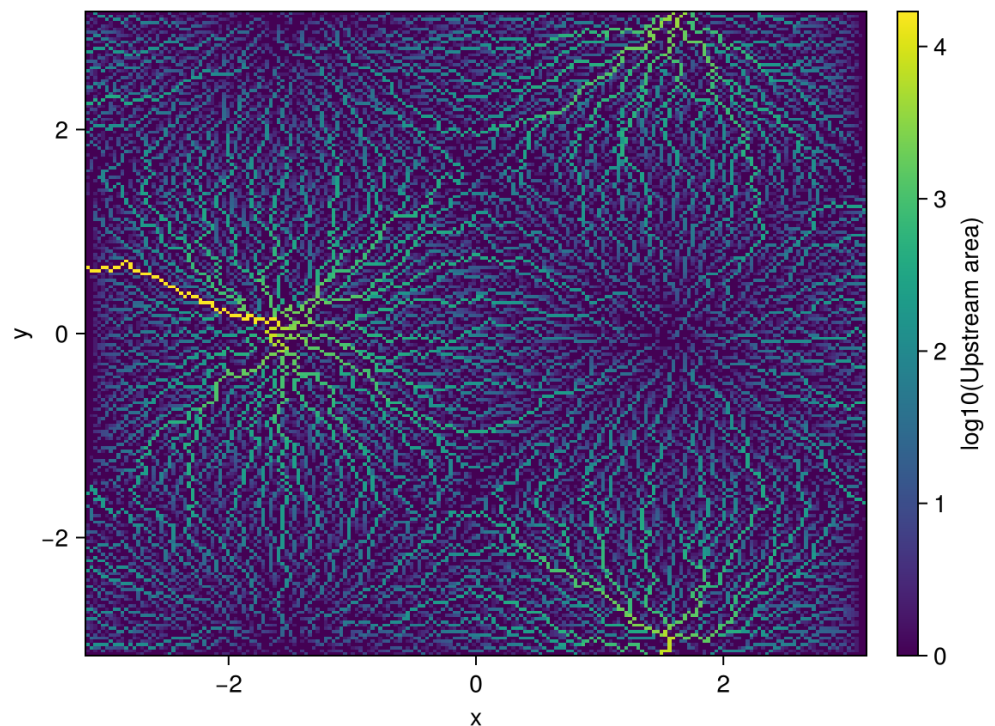


Figure 1: Upslope contributing area on a synthetic DEM from running the provided code example. Brighter colours indicate larger contributing areas, tracing the drainage network emerging from the synthetic topography.

Research impact

WhereTheWaterFlows.jl has been used in published glaciological research: Malczyk et al. (2023) used it to route subglacial melt water and constrain melt fluxes beneath an ice sheet from observations of active subglacial lake recharge. The study employed WWF's uncertainty quantification features. Delaney et al. (2023) built a 2D subglacial sediment transport model on top of WWF. Ogier et al. (2025) employed flow routing to analyse the likelihood of subglacial water pockets in Alpine glaciers, contributing to a new inventory for the Swiss Alps. Horgan et al. (2025) used flow routing to delineate the subglacial catchment of the Kamb Ice Stream. Washam et al. (2026) demonstrated that volcanic heat sources within the Kamb catchment likely contribute to enhanced subglacial melt. Ogier et al. (2026) used the package to assess the potential contribution of subglacial water reservoirs to the catastrophic 2024 La

102 Bérarde flood.

103 WWF is used for teaching in the lecture courses “Physics of Glaciers” at ETH Zurich
104 (Switzerland) and “Introduction to geoscientific programming” at Uni Mainz (Germany).
105 The software is archived on Zenodo (Werder et al., 2024) and has accumulated 20+ GitHub
106 stars and 5+ forks since its public release in 2019. It has four contributors and 200+ commits
107 across 20+ tagged releases.

108 AI usage disclosure

109 AI tools from Anthropic and OpenAI are used in the development of WWF. Most of the code
110 predates 2022, the advent of capable LLMs, and was thus written by hand. This also applies
111 to the Randomly and Subglacially submodules which were previously developed in a separate,
112 private repository and only now merged into WWF. Recently, code refactoring was helped
113 by LLMs. Much of the documentation, examples and benchmarks were generated with the
114 help of LLMs. The tools employed to date are Claude Sonnet 4.6 and GPT-5.3-Codex. All AI
115 generated content was reviewed by a human.

116 The first draft of this manuscript was created by Claude Sonnet 4.6. The tool was used for
117 initial text generation and structural organisation of the paper sections. The text was then
118 reviewed, heavily edited and rewritten by the two authors.

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